

Brand Category	Key Players	Impact Level	What's Happening
Memory Manufacturers	Samsung, SK Hynix, Micron	High (Positive)	Record margins as production shifts toward HBM and enterprise-grade DRAM
Premium Device Makers	Apple, Samsung (Mobile)	Moderate	Long-term contracts soften the blow, but RAM upgrades stall
PC & Laptop OEMs	Dell, HP, Lenovo, ASUS	High (Negative)	15–20% price increases driven by higher memory BOM costs
Budget & Mid-Range Brands	Xiaomi, Realme, Transsion	Critical	Thin margins force price hikes or downgraded RAM configurations
AI Infrastructure Providers	Nvidia, Microsoft, Google, Amazon	Priority Access	Securing future supply through long-term, high-value contracts

term supply agreements, and bargaining power within the supply chain.

Memory manufacturers are the clear winners. Samsung, SK Hynix, and Micron are posting record margins by prioritising AI-focused memory. DRAM contract prices rose sharply through 2025, with year-on-year increases exceeding 170% in some categories.

Premium device makers sit in a protected middle ground. Companies like Apple and Samsung's mobile division have long-term supply agreements and the balance sheets to secure memory 12 to 24 months in advance. They still feel the memory shortage, but they absorb it quietly through frozen RAM upgrades and slower price erosion rather than sudden shocks.

PC and laptop OEMs are far more exposed. Brands such as Dell, HP, Lenovo, and ASUS have already signalled 15–20 percent price increases. In their bill of materials, memory is too large a component to hide. The memory supply shortage shows up directly on the price tag.

The most vulnerable players are budget and mid-range brands operating on razor-thin margins. Companies like Xiaomi, Realme, and Transsion cannot easily absorb rising costs. For them, the memory supply shortage forces uncomfortable choices: raise prices in price-sensitive markets or quietly cut specifications.

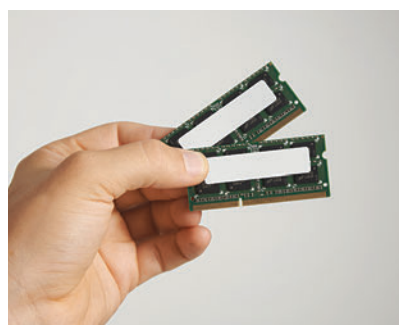
Meanwhile, AI infrastructure players sit at the front of the line. Companies like Nvidia, Microsoft, Google, Meta, and Amazon are securing large portions of future memory output through long-term agreements. Their

purchasing power allows them to outbid other customers, ensuring supply continuity even as shortages persist elsewhere in the market.

Categories facing price rise

The memory shortage does not hit all products equally. Some categories are absorbing far more than others.

High-performance DDR5 memory for gaming PCs is already showing signs of speculative behaviour. With fabs prioritising HBM, consumer DDR5 output has tightened. Enthusiasts are



fear-buying kits, and scalpers have returned to a market that felt boring.

Enterprise-grade NVMe SSDs are also under significant pressure. NAND flash prices have surged as manufacturers focus on high-density drives destined for AI training clusters. For data centres and professional workstations, the memory supply shortage means longer lead times and sharply higher procurement costs.

Mid-range smartphones are quietly changing too. To cope with the memory supply shortage, brands are freezing RAM configurations. Where 8 or 12 GB once became standard, 4 or

6 GB could reappear as a cost-control measure. Consumers may not notice immediately, but the slowdown in spec progression is real.

Graphics cards are feeling the squeeze due to VRAM availability. Mid-tier GPUs shipping with reduced memory? No longer far-fetched. When VRAM competes with AI accelerators for allocation, the memory shortage reshapes GPU roadmaps.

AI-powered PCs face a particularly awkward collision. Microsoft's Copilot+ guidelines require at least 16 GB of RAM, but the memory supply shortage makes that baseline expensive. Entry-level AI PCs are becoming premium products, undermining the very narrative they were meant to support.

Consumer memory lost

One of the clearest signals of how the memory supply shortage is reshaping priorities came from Micron stepping back from the Crucial consumer memory business. For years, Crucial RAM represented a reliable, accessible option for students, small businesses, and PC enthusiasts. Its exit was not driven by declining demand for consumer RAM, but by the opportunity cost of serving that market. AI-focused products offer higher margins, longer contracts and strategic importance to hyperscale customers. In comparison, consumer memory involves marketing costs, retail distribution and comparatively lower returns. **D**

Read more on Digit.in:
tinyurl.com/2zcj4wht